Lecture 24: The Cross Product

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Define the cross product of two vectors in three-dimensional space and describe its geometric significance.
- ☐ Compute the cross product using the determinant formula.
- ☐ Determine whether two nonzero vectors are parallel.
- ☐ Apply properties of the cross product in vector algebra.
- ☐ Calculate the area of a parallelogram and a triangle defined by two vectors using the magnitude of the cross product.
- ☐ Find a vector perpendicular to a given plane defined by two non-parallel vectors.
- ☐ Evaluate the scalar triple product and interpret it as the volume of a parallelepiped.
- ☐ Determine whether vectors are coplanar using the scalar triple product.

Cross Product

Let $\boldsymbol{u}$ and $\boldsymbol{v}$ be two non-parallel vectors in space, then they determine a plane.

We select a unit vector $\boldsymbol{n}$ perpendicular to the plane by the right-hand rule.

We define the cross product $\boldsymbol{u} \times \boldsymbol{v}$ (read as $\boldsymbol{u}$ cross $\boldsymbol{v}$) be the vector

$$\boldsymbol{u} \times \boldsymbol{v} = (|\boldsymbol{u}||\boldsymbol{v}| \sin\theta)\boldsymbol{n}$$

Note that $\boldsymbol{u} \times \boldsymbol{v}$ is parallel to the vector $\boldsymbol{n}$, so it is orthogonal to both $\boldsymbol{u}$ and $\boldsymbol{v}$.

Cross Product

Parallel Vectors

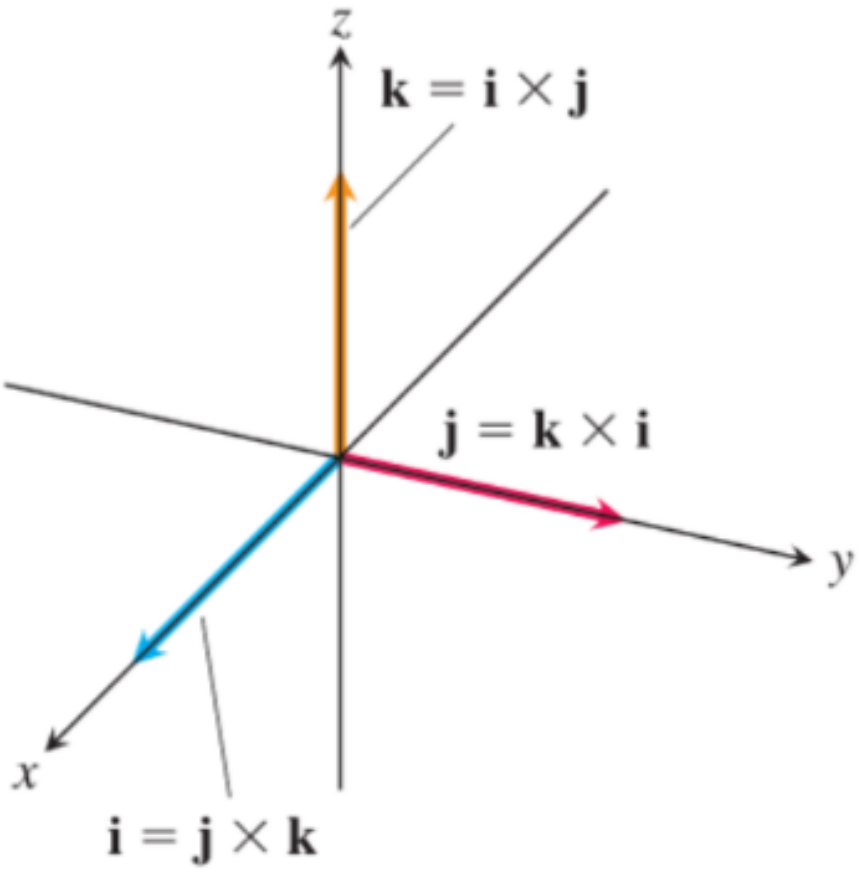
Nonzero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are parallel if and only if $\boldsymbol{u} \times \boldsymbol{v} = \boldsymbol{0}$.

Properties of the Cross Product

If $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{w}$ are any vectors and r, s are scalars, then

1. $(r\boldsymbol{u}) \times (s\boldsymbol{v}) = (rs)(\boldsymbol{u} \times \boldsymbol{v})$
2. $\boldsymbol{u} \times (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \times \boldsymbol{v} + \boldsymbol{u} \times \boldsymbol{w}$
3. $\boldsymbol{v} \times \boldsymbol{u} = -(\boldsymbol{u} \times \boldsymbol{v})$
4. $(\boldsymbol{v} + \boldsymbol{w}) \times \boldsymbol{u} = \boldsymbol{v} \times \boldsymbol{u} + \boldsymbol{w} \times \boldsymbol{u}$
5. $\boldsymbol{0} \times \boldsymbol{u} = \boldsymbol{0}$
6. $\boldsymbol{u} \times (\boldsymbol{v} \times \boldsymbol{w}) = (\boldsymbol{u} \cdot \boldsymbol{w})\boldsymbol{v} - (\boldsymbol{u} \cdot \boldsymbol{v})\boldsymbol{w}$

Cross Product



Area of a Parallelogram

$$|\boldsymbol{u} \times \boldsymbol{v}| = |\boldsymbol{u}||\boldsymbol{v}||\sin\theta||\boldsymbol{n}| = |\boldsymbol{u}||\boldsymbol{v}|\sin\theta$$

Determinant Formula for $\boldsymbol{u} \times \boldsymbol{v}$

Let $\boldsymbol{u} = u_1\boldsymbol{i} + u_2\boldsymbol{j} + u_3\boldsymbol{k}$ and $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$.

$\boldsymbol{u} \times \boldsymbol{v} =$

Calculating Cross Product

Example: Find $\boldsymbol{u} \times \boldsymbol{v}$ and $\boldsymbol{v} \times \boldsymbol{u}$ if $\boldsymbol{u} = 2\boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k}$ and $\boldsymbol{v} = -4\boldsymbol{i} + 3\boldsymbol{j} + \boldsymbol{k}$.

Calculating Cross Product

Example: Find a vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

Calculating Cross Product

Example: Find the area of the triangle with vertices $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

Calculating Cross Product

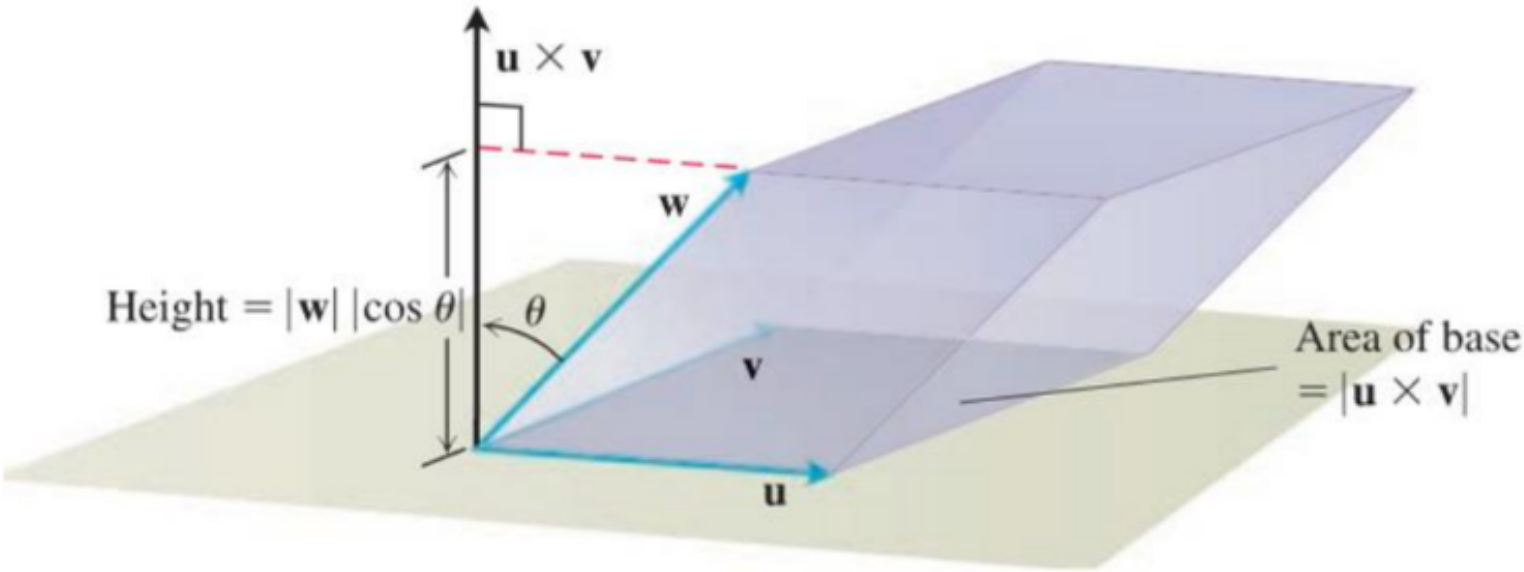
Example: Find a unit vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

Calculating Cross Product

Example: Show that $|\boldsymbol{u} \cdot \boldsymbol{v}|^2 + |\boldsymbol{u} \times \boldsymbol{v}|^2 = |\boldsymbol{u}|^2 |\boldsymbol{v}|^2$.

Triple Scalar Product

The product $(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w}$ is called the triple scalar product (also known as box product) of $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$.



Volume = area of base · height

Properties of Triple Scalar Product

$(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w} = (\boldsymbol{v} \times \boldsymbol{w}) \cdot \boldsymbol{u} = (\boldsymbol{w} \times \boldsymbol{u}) \cdot \boldsymbol{v}$.

$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ are coplanar iff $(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w} = 0$

Triple Scalar Product as Determinant

Let $\boldsymbol{u} = u_1\boldsymbol{i} + u_2\boldsymbol{j} + u_3\boldsymbol{k}$, $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$ and $\boldsymbol{w} = w_1\boldsymbol{i} + w_2\boldsymbol{j} + w_3\boldsymbol{k}$.

$(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w} =$

Calculating Triple Scalar Product

Example: Find the volume of the parallelepiped determined by $\boldsymbol{u} = \boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$, $\boldsymbol{v} = -2\boldsymbol{i} + 3\boldsymbol{k}$ and $\boldsymbol{w} = 7\boldsymbol{j} - 4\boldsymbol{k}$.

Calculating Triple Scalar Product

Example: Show that $\boldsymbol{u} = \boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k}$, $\boldsymbol{v} = \boldsymbol{i} + 2\boldsymbol{j} + 3\boldsymbol{k}$ and $\boldsymbol{w} = 2\boldsymbol{i} + 3\boldsymbol{j} + 4\boldsymbol{k}$ are coplanar.